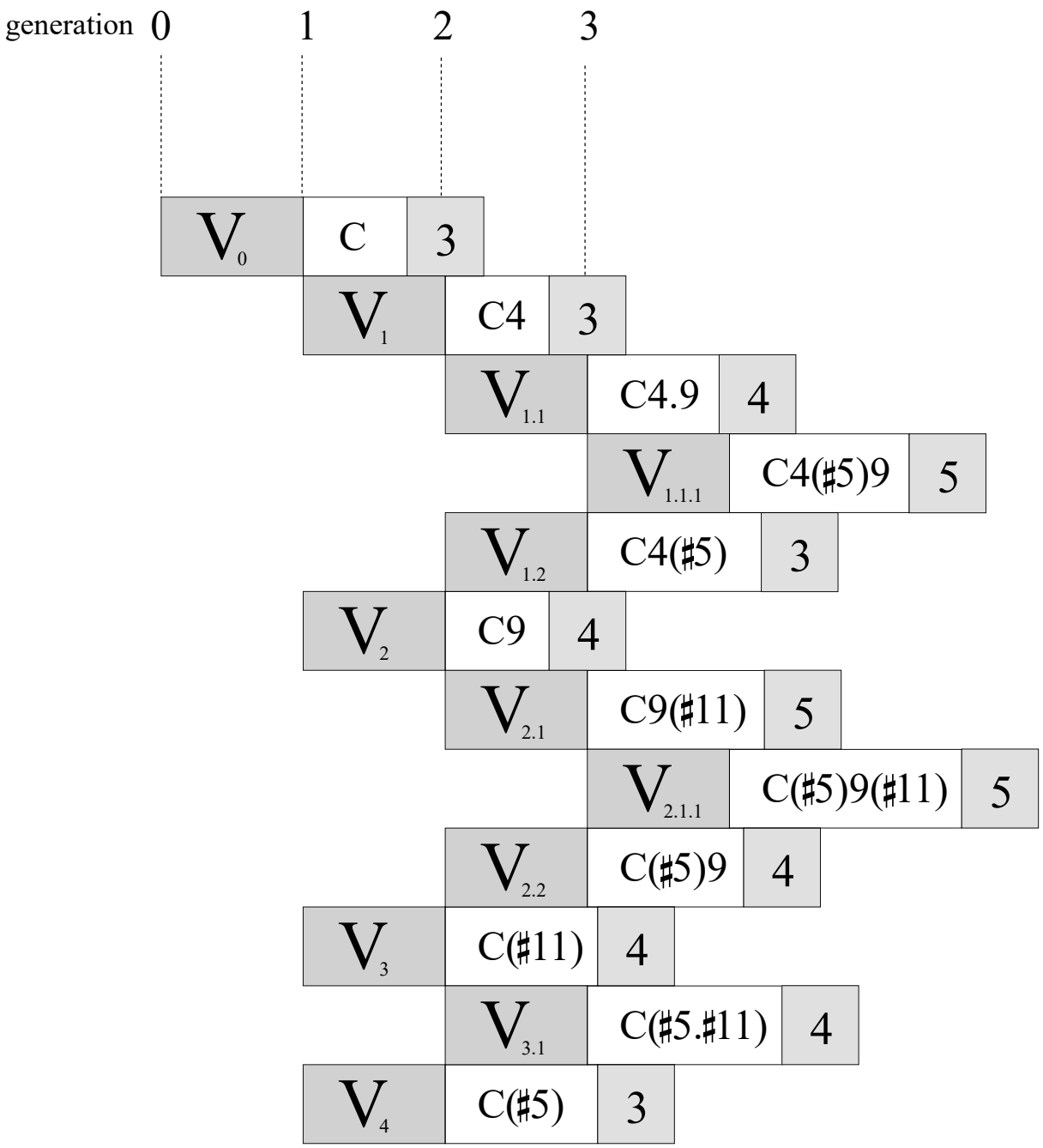
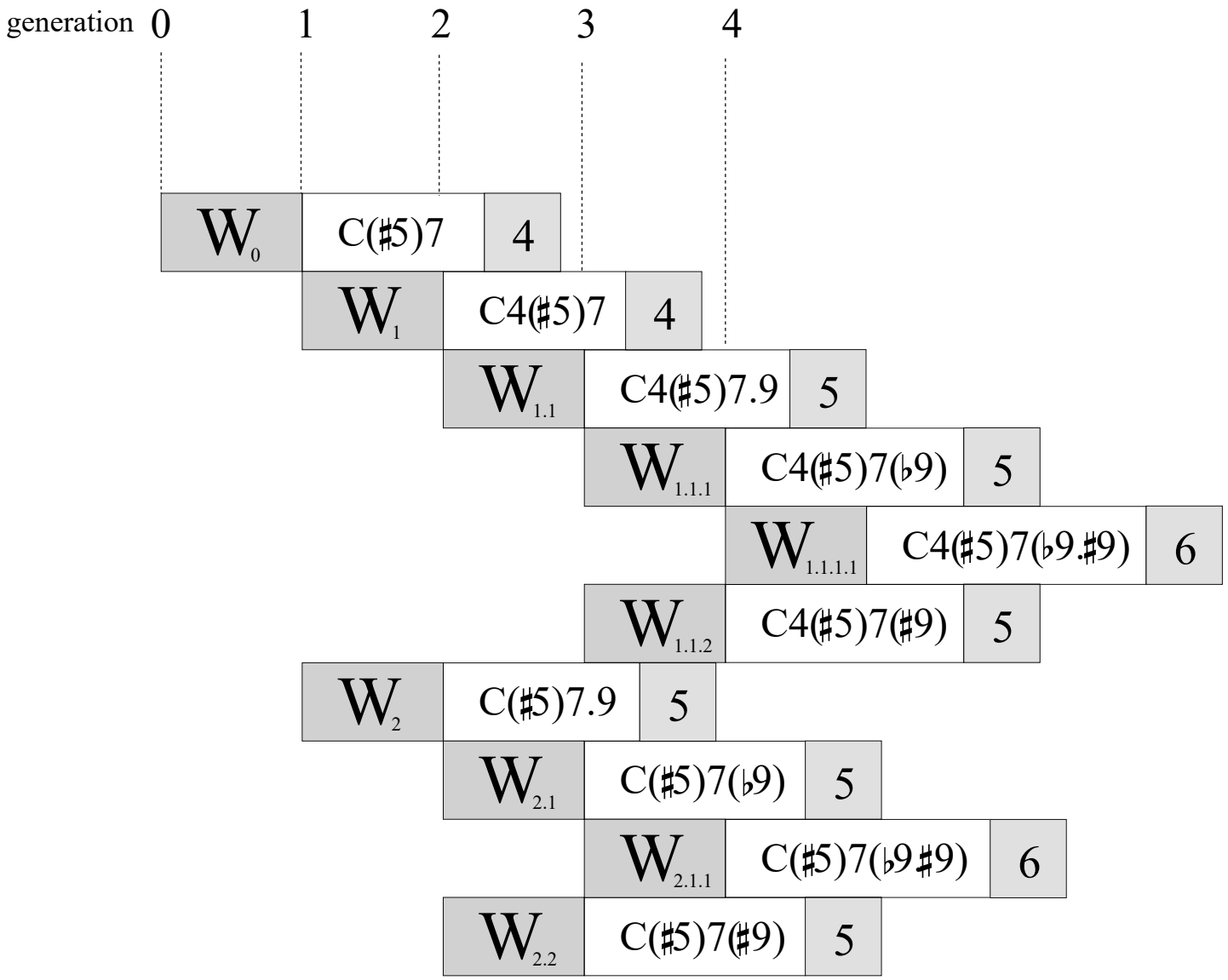
Complete Genealogy of the Protochords

This document presents the complete genealogy of the 10 proto-chords, resulting in 161 chord types. Each figure corresponds to the genealogy of a proto-chord, and each block shows the genealogical notation, the usual alphanumeric chord notation (using C as the root), and the number of notes in that chord. The vertical bars indicate the generation of each chord.

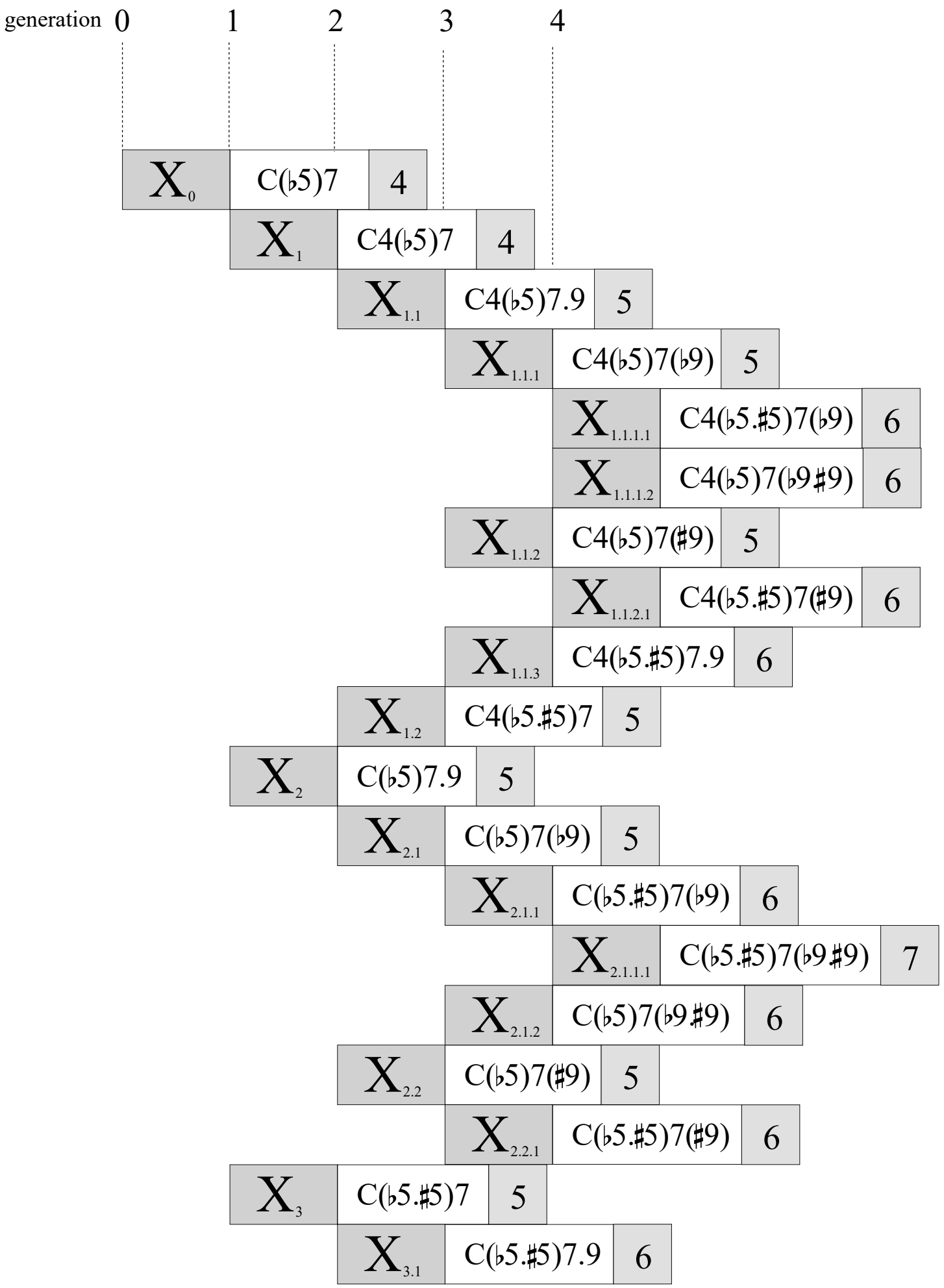
V (major triad)



W (seventh chord with augmented fifth)



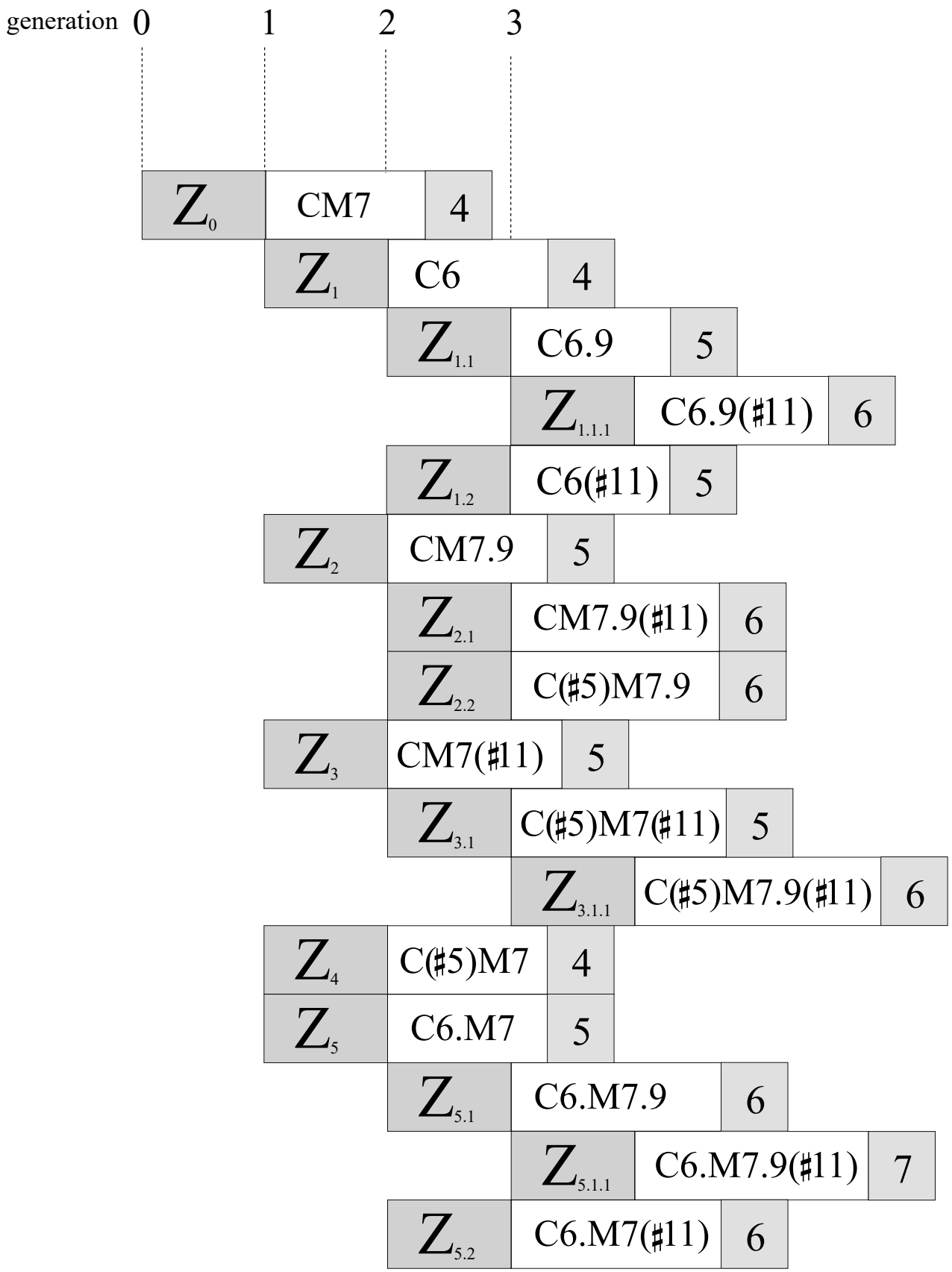
X (seventh chord with diminished fifth)



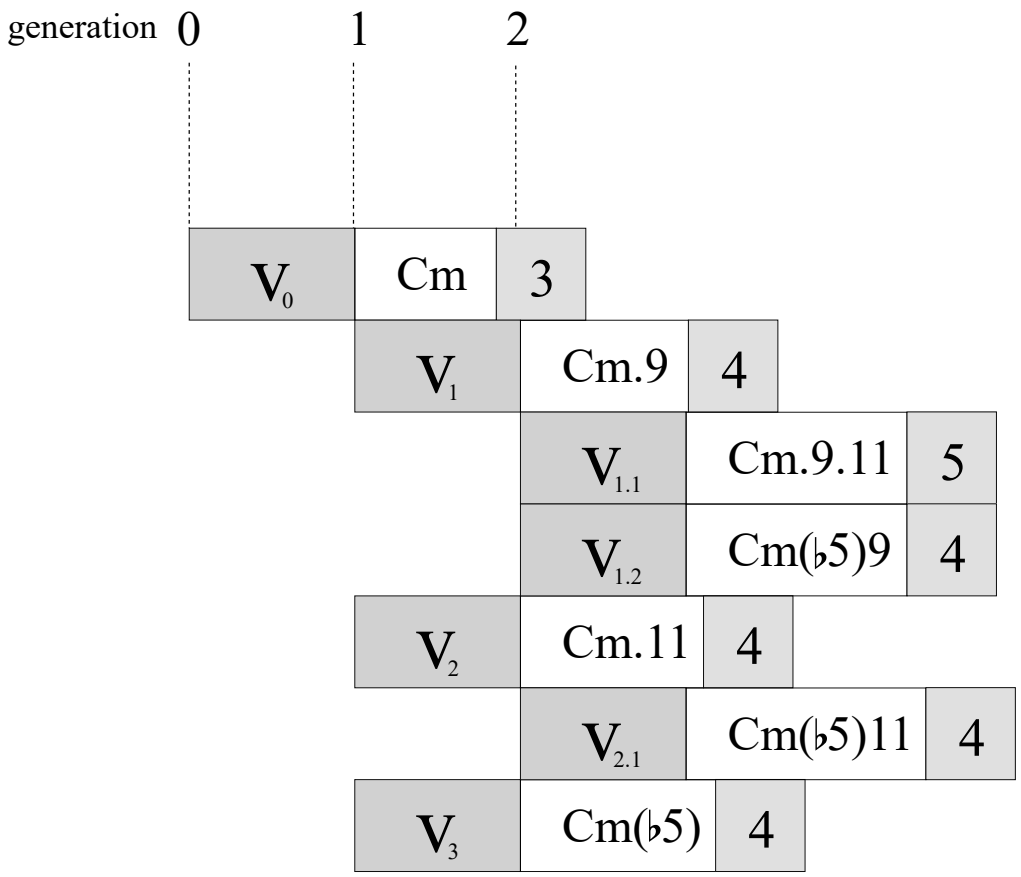
Y (seventh chord)

genertion	0	1	2	3	4	5
	Y_0	C7	4			
		Y_1	C4.7	4		
			$Y_{1.1}$	C4.7.9	5	
				$Y_{1.1.1}$	C4.7.9(♯11)	6
					$Y_{1.1.1.1}$	C4.7.9(♯11)13	7
						$Y_{1.1.1.1.1}$	C4.7.9(♯11.♭13)	7
						$Y_{1.1.1.1.2}$	C4.7(♭9.♯11.♭13)	7
						$Y_{1.1.1.1.3}$	C4.7(♯9.♯11.♭13)	7
					$Y_{1.1.1.2}$	C4.7(♭9.♯11)	6
						$Y_{1.1.1.2.1}$	C4.7(♭9.♯11)13	7
						$Y_{1.1.1.2.2}$	C4.7(♭9.♯9.♯11)13	8
					$Y_{1.1.1.3}$	C4.7(♯9.♯11)	6
						$Y_{1.1.1.3.1}$	C4.7(♯9.♯11)13	7
				$Y_{1.1.2}$	C4.7.9.13	6
					$Y_{1.1.2.1}$	C4.7.9(♭13)	6
						$Y_{1.1.2.1.1}$	C4.7(♭9.♭13)	6
						$Y_{1.1.2.1.2}$	C4.7(♯9.♭13)	6
					$Y_{1.1.2.2}$	C4.7(♭9)13	6
					$Y_{1.1.2.3}$	C4.7(♯9)13	6
				$Y_{1.1.3}$	C4.7(♭9)	5
					$Y_{1.1.3.1}$	C4.7(♭9.♯9)	6
				$Y_{1.1.4}$	C4.7(♯9)	5
			$Y_{1.2}$	C4.7(♯11)	5
				$Y_{1.2.1}$	C4.7(♯11)13	6
					$Y_{1.2.1.1}$	C4.7(♯11.♭13)	6
			$Y_{1.3}$	C4.7.13	5
				$Y_{1.3.1}$	C4.7(♭13)	5
		Y_2	C7.9	5
			$Y_{2.1}$	C7.9(♯11)	6
				$Y_{2.1.1}$	C7.9(♯11)13	7
					$Y_{2.1.1.1}$	C7(♭9.♯11)13	7
					$Y_{2.1.1.2}$	C7(♯9.♯11)13	7
					$Y_{2.1.1.3}$	C7.9(♯11.♭13)	7
				$Y_{2.1.2}$	C7(♭9.♯11)	6
					$Y_{2.1.2.1}$	C7(♭9.♯9.♯11)	7
				$Y_{2.1.3}$	C7(♯9.♯11)	6
					$Y_{2.1.3}$	C7(♯9.♯11.♭13)	7
			$Y_{2.2}$	C7.9.13	6
				$Y_{2.2.1}$	C7(♭9)13	6
					$Y_{2.2.1.1}$	C7(♭9.♭13)	6
					$Y_{2.2.1.2}$	C7(♭9.♯9)13	7
				$Y_{2.2.2}$	C7(♯9)13	6
					$Y_{2.2.2}$	C7(♯9)13	6
						$Y_{2.2.2.1}$	C7(♯9.♭13)	6
			$Y_{2.3}$	C7(♭9)	5
				$Y_{2.3.1}$	C7(♭9.♯9)	6
			$Y_{2.4}$	C7(♯9)	5
		Y_3	C7(♯11)	5
			$Y_{3.1}$	C7(♯11)13	6
				$Y_{3.1.1}$	C7(♯11.♭13)	6
		Y_4	C7.13	5
			$Y_{4.1}$	C7(♭13)	5

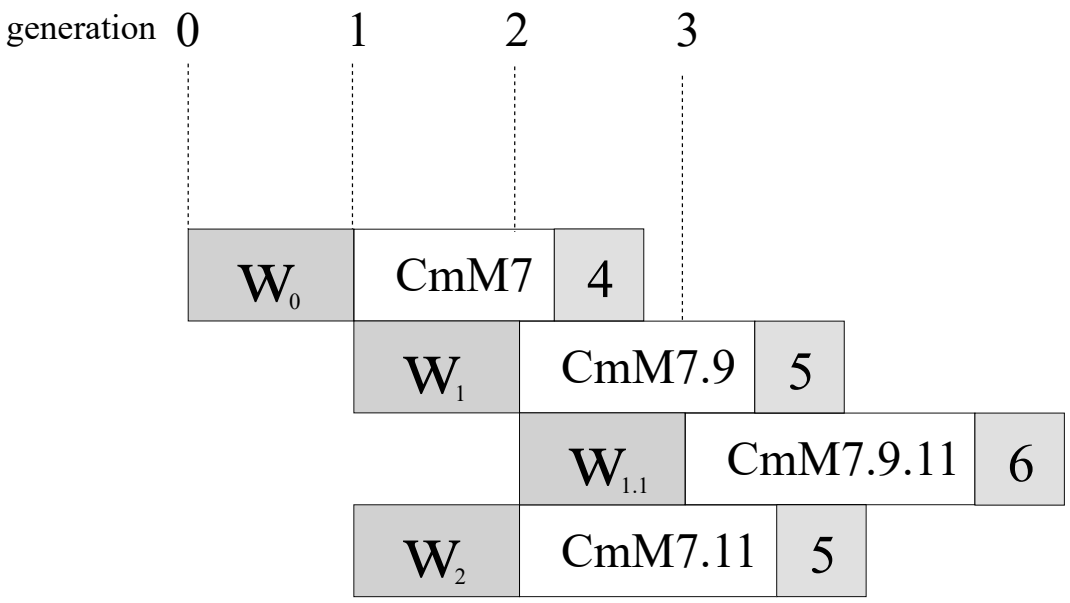
Z (major seventh chord)



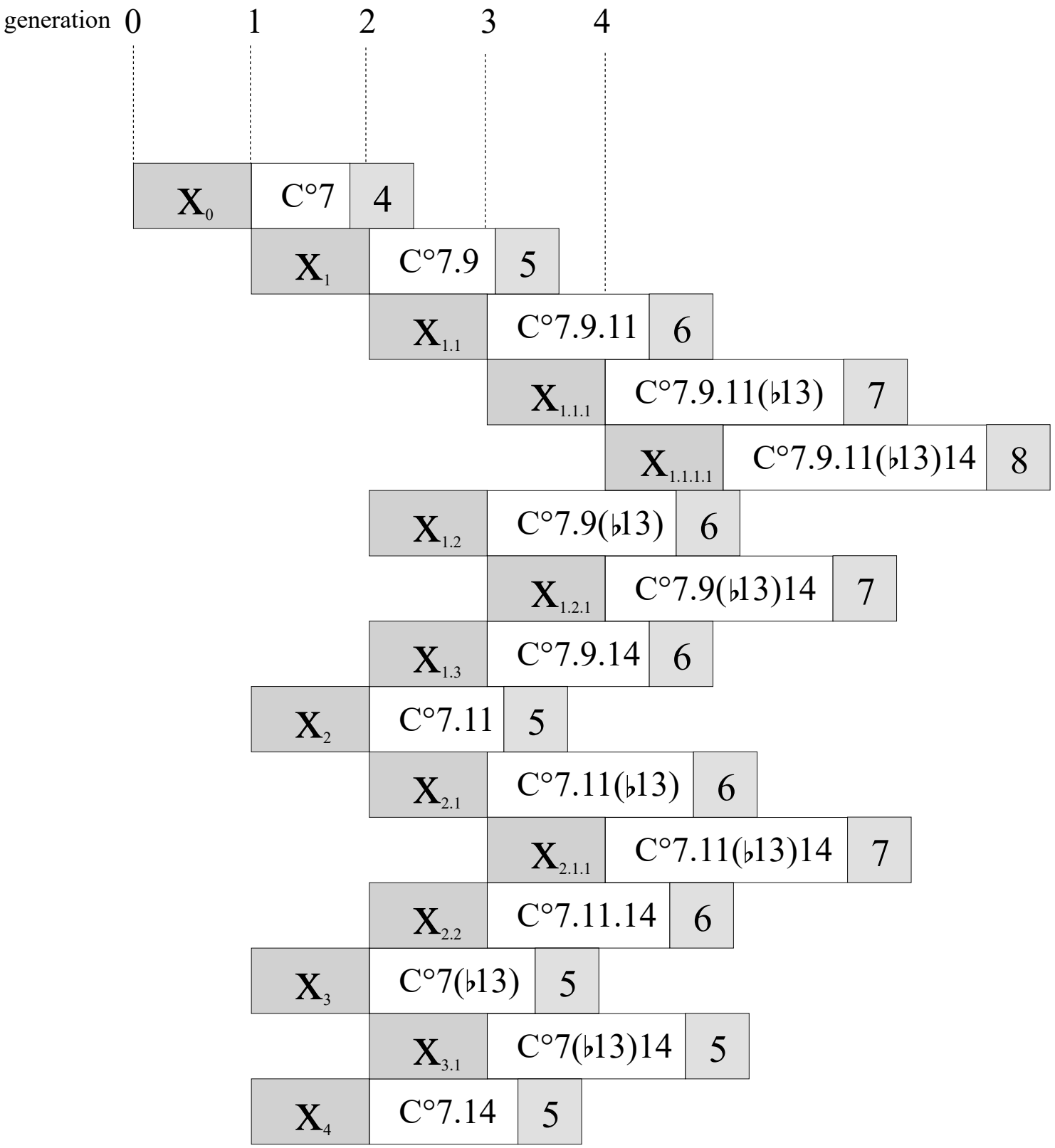
v (minor triad)



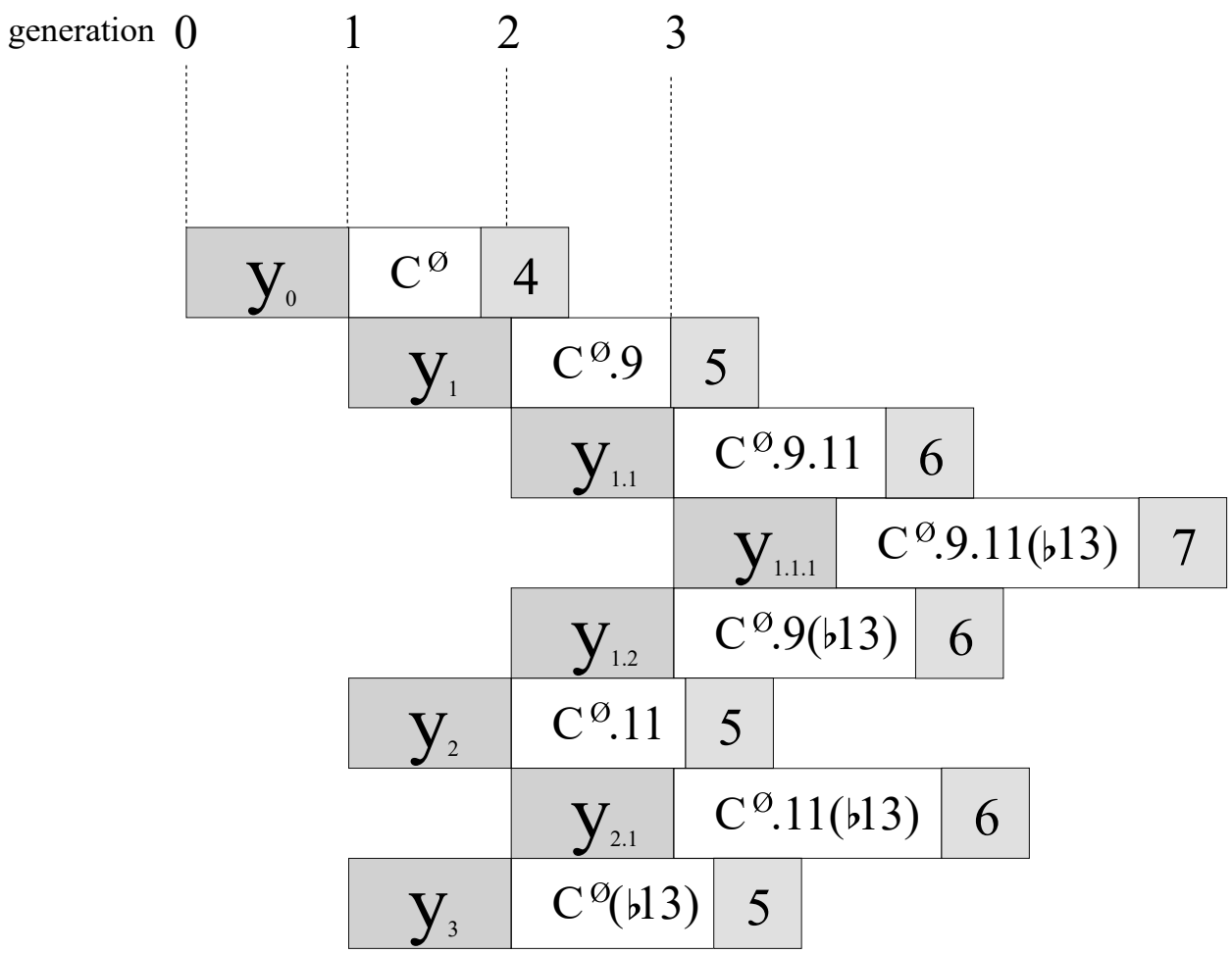
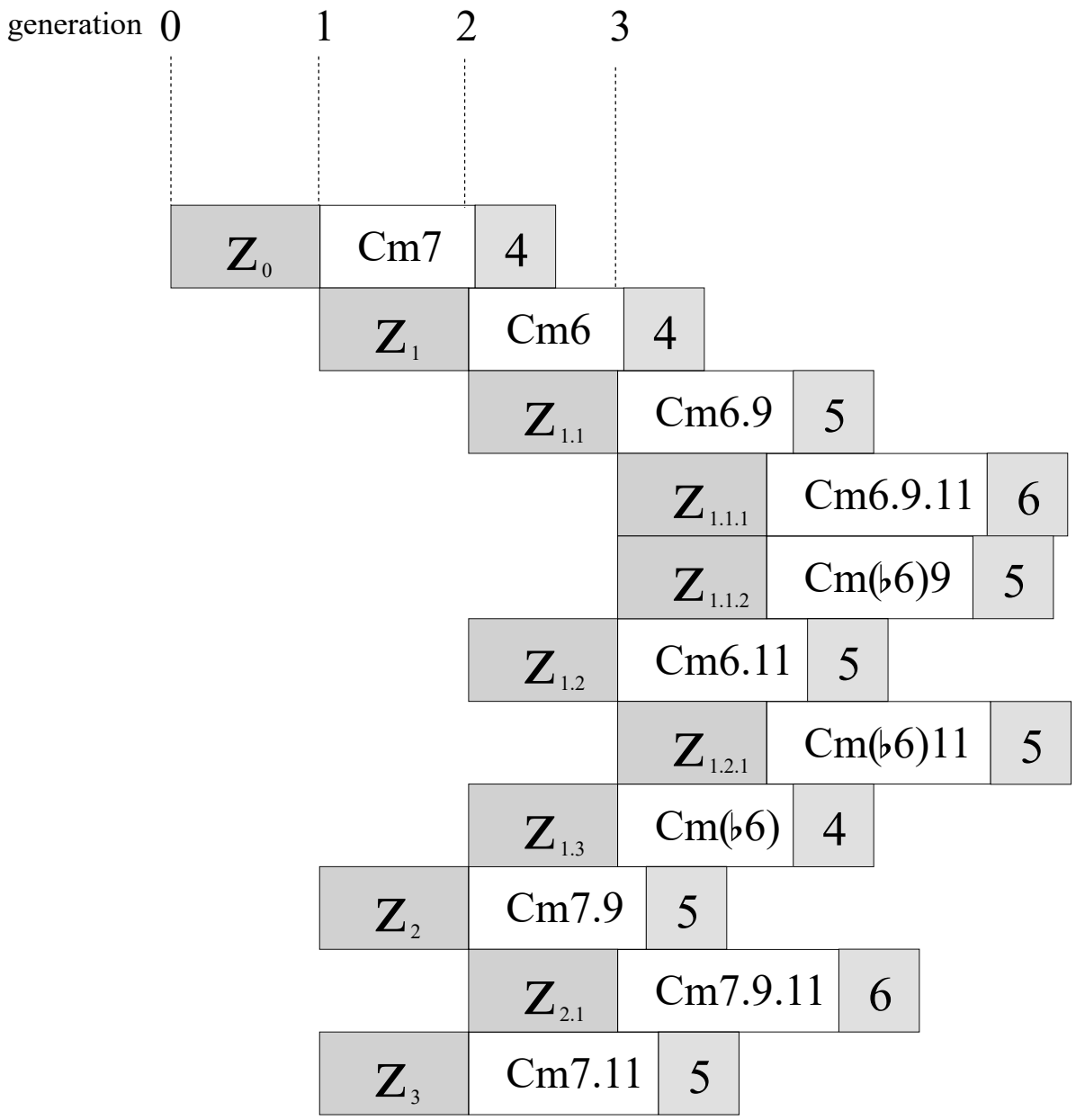
v (minor triad with major seventh)



x (diminished triad with seventh)



y (half-diminished chord)



z (minor triad with seventh)